

Public Realm Cost Calculation

Original Requirement (from Project Documentation)

Calculation of Public Realm Cost

This will be calculated based on the selected lines of Public Realm typologies as follows: District Park, Neighborhood Park, Local Park, Pocket Park, and Buffer Landscapes and Trails

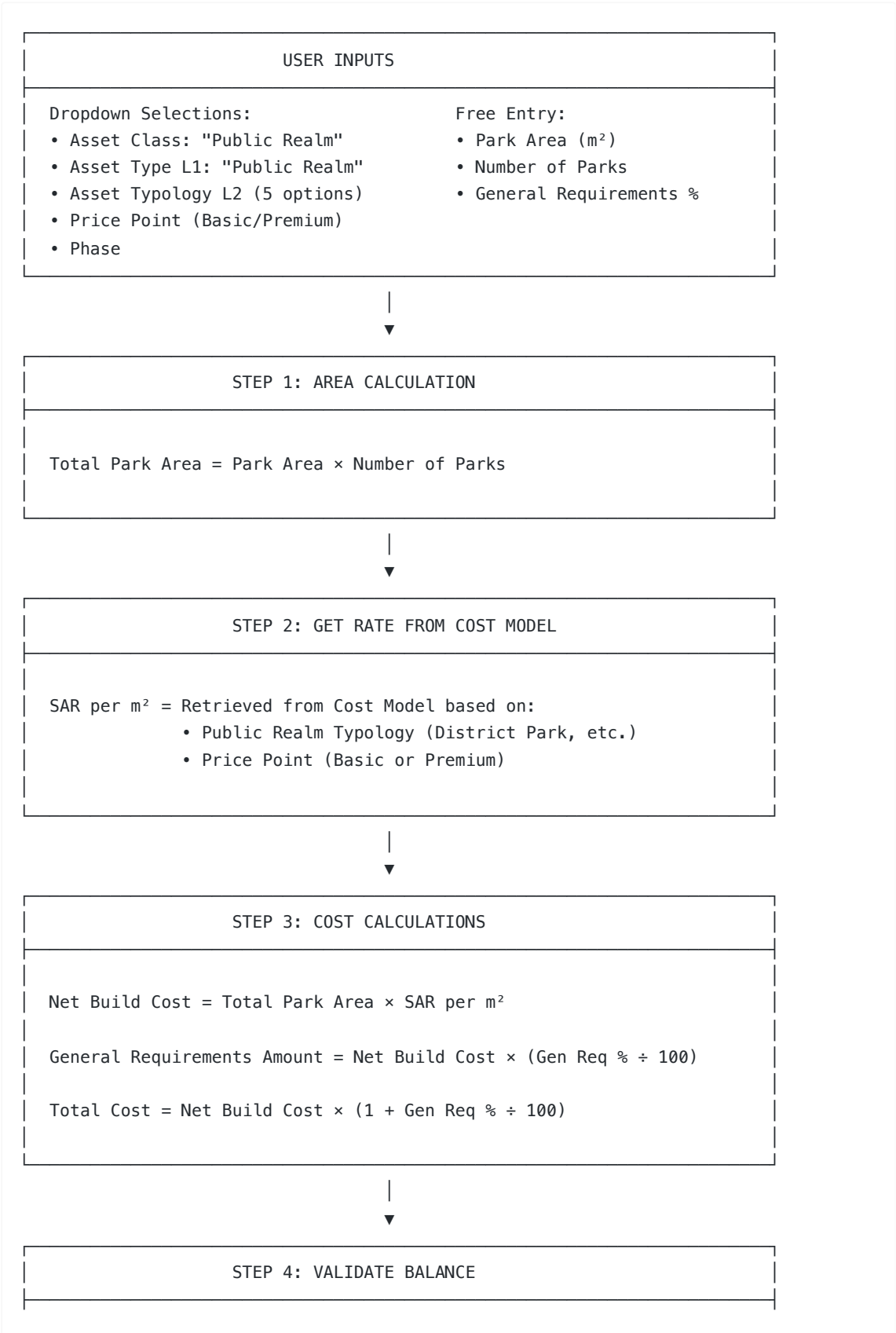
- 1. Area of each type will be multiplied by the rates inserted in the configuration based on each price Point*
- 2. Remaining/Balance of external area (GLA - total building assets plot areas - public realm areas) will be calculated automatically and it will give error if negative.*

Public Realm Typologies

The platform supports five types of public realm spaces:

PUBLIC REALM TYPOLOGIES
DISTRICT PARK Large community parks serving multiple neighborhoods
NEIGHBORHOOD PARK Medium-sized parks serving a single neighborhood
LOCAL PARK Smaller parks serving immediate local area
POCKET PARK Small urban parks in compact spaces
BUFFER LANDSCAPES AND TRAILS Linear parks, walking trails, green buffers

Calculation Flow Diagram



Balance External Area = GLA - Building Plot Areas - Public Realm Areas

└─ If Balance ≥ 0 : Valid allocation

└─ If Balance < 0: ERROR - Areas exceed available land



OUTPUT

- Total Park Area (m²)
- Net Build Cost (SAR)
- General Requirements Amount (SAR)
- Total Cost (SAR)
- Balance External Area (m²) with validation

Example Calculation

Input:

- Typology: Neighborhood Park
- Price Point: Premium
- Park Area: 5,000 m²
- Number of Parks: 3
- General Requirements: 10%
- SAR per m² (from Cost Model): 150 SAR

Step 1 - Area Calculation:

- Total Park Area = 5,000 × 3 = **15,000 m²**

Step 2 - Cost Calculation:

- Net Build Cost = 15,000 × 150 = **2,250,000 SAR**
- General Requirements = 2,250,000 × 0.10 = **225,000 SAR**
- Total Cost = 2,250,000 + 225,000 = **2,475,000 SAR**

Balance External Area Validation

The platform continuously validates that all allocated areas fit within the Gross Land Area:

BALANCE CALCULATION

Gross Land Area (from Masterplan)

└─ Minus: Total Building Plot Areas

Minus: Total Public Realm Areas
= Balance External Area

Example:

GLA: 500,000 m²

Building Plot Areas: 350,000 m²

Public Realm Areas: 80,000 m²

Balance: 500,000 - 350,000 - 80,000 = 70,000 m²

This 70,000 m² is available for roads, utilities, etc.

Error Condition

If the Balance External Area becomes negative, the platform shows an error:

"Total allocated areas exceed Gross Land Area by X m². Please reduce Building Assets plot areas or Public Realm areas."

Warning Condition

If the Balance External Area is less than 5% of the GLA, the platform shows a warning:

"Balance External Area is only X% of Gross Land Area. Consider if this is sufficient for roads, utilities, and other infrastructure."

Price Points

Public realm costs vary by price point:

Price Point	Description	Typical Use
Basic	Standard landscaping and amenities	Economy developments
Premium	High-quality materials and features	Premium developments

The rate per m² is different for each price point and is configured in the Cost Model.

Aggregated Totals

When multiple public realm entries are added, the platform calculates:

- **Total Public Realm Area:** Sum of all park areas
- **Total Public Realm Cost:** Sum of all park costs
- **Breakdown by Typology:** Area and cost for each park type